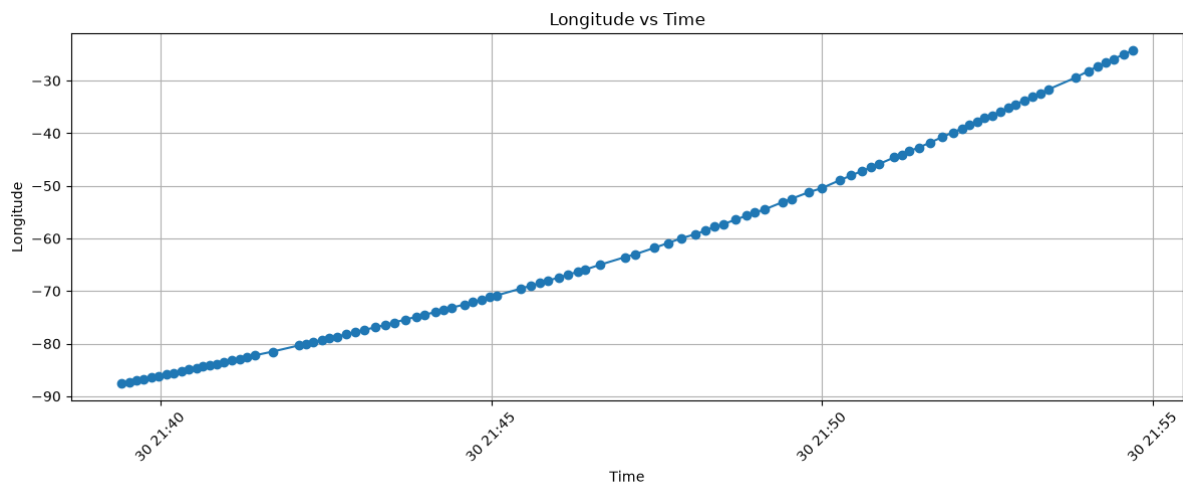
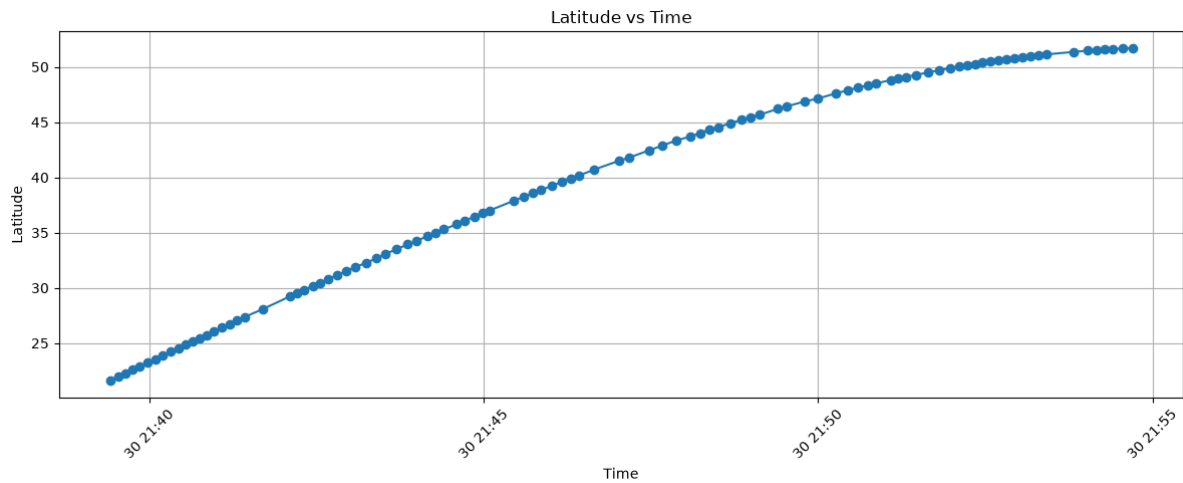


ASSIGNMENT REPORT

**Analysing and Interpreting Real-Time IoT
Data**

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Interpretation of Results

1. How did the ISS location change over time?

The International Space Station (ISS) showed a continuous change in its position throughout the observation period. As data was collected every five seconds, both the latitude and longitude values changed steadily, indicating that the ISS was continuously moving along its orbit around the Earth.

The latitude increased from approximately 21.58° to 51.70° , while the longitude increased from approximately -87.63° to -24.21° during the data collection period.

2. Did the latitude and longitude show continuous movement?

Yes. Both the latitude and longitude exhibited smooth and continuous movement.

- The latitude graph shows a gradual increase over time without sudden jumps or irregular fluctuations. This indicates that the ISS was moving northward during the observation period.
- The longitude graph also shows a continuous increase from west to east (from approximately -88° to -24°). The absence of abrupt changes indicates stable orbital motion.

The smooth nature of both graphs confirms that the ISS follows a predictable orbital path.

3. Observations from the Graphs

Latitude Graph

- The latitude increased steadily throughout the observation period.
- The curve is smooth and continuous, indicating uniform motion.
- The rate of increase becomes slightly smaller near the end, causing the graph to flatten slightly.
- No missing values or sudden spikes were observed.

Longitude Graph

- Longitude increased continuously from approximately -88° to -24° .
- The graph displays a smooth upward trend throughout the observation.
- There are no abrupt discontinuities or unusual fluctuations.
- The ISS maintained a consistent eastward movement during data collection.

Statistical Summary

Latitude

- Maximum Latitude: 51.70°
- Minimum Latitude: 21.58°
- Average Latitude: 39.18°

Longitude

- Maximum Longitude: -24.21°
- Minimum Longitude: -87.63°
- Average Longitude: -61.63°

Conclusion

This project successfully retrieved real-time ISS location data using a public API and analyzed its movement over time. A total of 100 consecutive observations were collected at 5-second intervals and stored in a CSV file.

The generated graphs demonstrate that both latitude and longitude changed continuously throughout the observation period, confirming the continuous orbital motion of the ISS. The smooth trends in the graphs indicate stable movement without interruptions or sudden positional changes. The statistical analysis further summarizes the range and average position of the ISS during the data collection period.

Overall, the project demonstrates the use of Python for real-time data acquisition, data storage, statistical analysis, and data visualization, providing a clear understanding of the ISS's movement over time.